

DevOps Basics

Cognizant Digital Nurture 5.0 – Deep Skilling Program

Week 7 – DevOps & Cloud

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1. Introduction to DevOps

Modern software development demands faster delivery, high-quality software, and continuous customer feedback. Traditionally, development and operations teams worked independently, leading to communication gaps, delayed releases, and deployment failures.

DevOps bridges this gap by combining development and operations practices into a unified workflow, enabling organizations to deliver software rapidly and reliably.

2. What is DevOps?

DevOps is a software development methodology that integrates **Development (Dev)** and **Operations (Ops)** to improve collaboration, automate workflows, and accelerate software delivery.

Definition

DevOps is a culture and set of practices that combines software development and IT operations to enable faster, more reliable software delivery.

3. Why DevOps?

Organizations adopt DevOps to:

- Deliver software faster
 - Improve collaboration
 - Automate repetitive tasks
 - Reduce deployment failures
 - Increase software quality
 - Improve customer satisfaction
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4. Problems Before DevOps

Traditional software development faced several challenges:

- Poor communication between teams
- Manual deployments
- Long release cycles
- High deployment failure rates
- Slow bug fixes
- Inconsistent environments

DevOps addresses these issues through automation, collaboration, and continuous monitoring.

5. DevOps Lifecycle

The DevOps lifecycle consists of continuous phases:

Plan



Develop



Build



Test



Release



Deploy



Operate



Monitor



Feedback

Lifecycle Phases

Phase	Description
Plan	Define project requirements
Develop	Write application code
Build	Compile and package applications
Test	Perform automated testing
Release	Prepare software for deployment
Deploy	Deploy to production
Operate	Manage running applications
Monitor	Track performance and issues
Feedback	Improve future releases

6. DevOps Culture

DevOps is based on:

- Collaboration
- Automation
- Continuous Learning

- Continuous Improvement
 - Shared Responsibility
 - Fast Feedback
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7. Continuous Integration (CI)

Continuous Integration is the practice of automatically integrating code changes into a shared repository multiple times a day.

Benefits

- Early bug detection
 - Faster builds
 - Better code quality
 - Reduced integration issues
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8. Continuous Delivery (CD)

Continuous Delivery ensures that software is always ready for deployment.

Every successful build passes through automated testing before being prepared for production.

9. Continuous Deployment

Continuous Deployment automatically deploys every successfully tested build to production without manual intervention.

Difference:

- **Continuous Delivery:** Manual approval before deployment.
 - **Continuous Deployment:** Fully automated deployment.
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10. DevOps Tools

Category	Popular Tools
Version Control	Git, GitHub
Build Tool	Maven, Gradle
CI/CD	Jenkins, GitHub Actions, GitLab CI
Containerization	Docker

Category	Popular Tools
Container Orchestration	Kubernetes
Monitoring	Prometheus, Grafana
Configuration Management	Ansible, Puppet, Chef
Cloud Platforms	AWS, Azure, Google Cloud

11. Benefits of DevOps

- Faster software delivery
 - Better collaboration
 - Improved software quality
 - Automated testing
 - Reduced downtime
 - Faster recovery
 - Continuous monitoring
 - Increased productivity
 - Scalability
 - Reliable deployments
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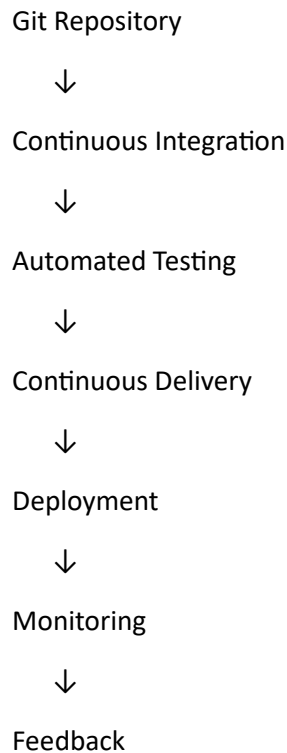
12. DevOps Best Practices

- Automate repetitive tasks
 - Use version control
 - Perform continuous testing
 - Monitor application performance
 - Use Infrastructure as Code (IaC)
 - Keep deployments small and frequent
 - Implement security early (DevSecOps)
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13. DevOps Workflow

Developer





14. DevOps vs Agile

Agile	DevOps
Focuses on software development	Focuses on development and operations
Faster development	Faster delivery
Ends after software development	Continues through deployment and monitoring
Team collaboration	Organization-wide collaboration

16. Conclusion

DevOps is a modern software development approach that promotes collaboration, automation, and continuous improvement throughout the software development lifecycle. By integrating development and operations practices, organizations can deliver high-quality software faster, reduce deployment risks, and improve overall operational efficiency. DevOps has become a fundamental practice in cloud-native application development and modern CI/CD workflows.